

Nicolas Rohart

Forestry - Design - Robotic Fabrication - Construction

From standing tree to finished structure.

I operate across the entire timber value chain myself - forest management and harvesting, computational design, robotic shaping of curved timber, and on-site construction - including the digital workflows that connect each stage.

BASE

Limousin, France

EXPERIENCE

16 years, full chain

LOOKING FOR

R&D partnership abroad

OPEN TO

Worldwide

THE CHAIN - FIVE STAGES, ONE HAND THROUGHOUT

01 Forest & Silviculture

Independent forest management, felling and sale, aimed at resilient long-term ecosystems rather than short-term extraction. Developer of "Bois Suivis" - a field-verifiable tracked-timber protocol that goes beyond declared certification schemes.

02 Harvest & Extraction

Felling and skidding done hands-on, in direct feedback with the rest of the chain. Built a custom skidding vehicle from a repurposed 4x4 fire-service chassis - bodywork, crane, structural welding - individually type-approved for road use.

03 Design

Custom timber structures designed from real, round, naturally curved trunks, fitted to each client's brief and site. Rhino 3D for modeling. TwinMotion for renders composited onto photogrammetry scans of the existing sites - near-photorealistic propositions that clients can already picture themselves in.

04 Robotic Shaping

Parametric Grasshopper / KukaPRC workflows generating toolpaths for a 7-axis KUKA robot that follows the tree's actual curvature rather than forcing it into a rectilinear beam. Dedicated 3D scanner: a 3 m trunk becomes a 600,000-point cloud in 3.5 minutes. Carried out at La Manufacture du Châtaignier with mentor Ludovic Mallegol.

05 On-Site Building

Full on-site installation, from site preparation and foundations to the final timber frame - combining traditional timber craftsmanship with the precision of robotic fabrication.

NETWORK

- IBOIS (EPFL Lausanne)
- GC2D (Égletons)
- IUT du Limousin (Limoges).

WHAT I'M LOOKING FOR

A research team, studio, or workshop working on wood, robotics, or both - bringing 16 years of hands-on chain experience into a lab or production environment as developer or trainer. Based in France, open to relocating.

Robotic fabrication

Timber engineering

Grasshopper / KukaPRC

Forestry & silviculture

Technical training

PUBLICATION

"The tree dictates the shape: reversing the industrial paradigm in rural areas"

L. Mallegol, N. Rohart, N. Champroy - Construction Robotics (Springer), 2026. Scan-to-trajectory robotic shaping of natural curved timber, developed at Coop Monts et Barrages.

TECHNICAL PROFILE

FORESTRY

Forest assessment

Silviculture

Timber harvesting

DIGITAL FABRICATION

Parametric design → toolpath

(Rhino / Grasshopper / KukaPRC)

Commercial renders

(MeshRoom / TwinMotion)

CONSTRUCTION

Design

Robot operation

Site implementation